Now, Chain rule for conditional Probability

$$P(C_k \mid x) = \frac{P(x \mid C_k) \cdot P(C_k)}{P(x)}$$

remain common so;

$$P(C_k \mid x) = P(x \mid C_k) \cdot P(C_k)$$

$$= \frac{P(x \cap C_k)}{P(C_k)} \cdot P(C_k)$$

$$= P(x \cap C_k)$$

$$= P(x, C_k)$$

replacing $\cap$ by comma ;

$$= P(\boxed{x_1}, \boxed{x_2, x_3, \ldots x_n, C_k})$$

~~let A ∩ B~~

~~P(A~~

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$= P(x_1, x_2, x_3, x_4, \ldots x_n, C_k)$$

$$= \underbrace{P(x_1 \mid x_2, x_3, \ldots, x_n, C_k)}_{a} \cdot P(x_2, x_3, \ldots x_n, C_k)$$

or, $P(C_k \mid x) = a \cdot P(\boxed{x_2}, \boxed{x_3, \ldots x_n, C_k})$

$$= a \cdot \underbrace{P(x_2 \mid x_3, x_4, \ldots x_n, C_k)}_{b} \cdot P(x_3, x_4, \ldots x_n, C_k)$$

$$= ab\, P(x_3, x_4, \ldots, x_n, C_k)$$

$$= \cdot \cancel{P(x_n \mid C_k) \cdot P(C_k)}$$

$$= ab\, P(x_n \mid C_k) \cdot P(C_k)$$

$$\therefore P(C_k \mid x) = P(x_1 \mid x_2, x_3 \ldots, x_n, C_k) \cdot P(x_2 \mid x_3, \ldots x_n, C_k) \ldots P(x_{n-1} \mid x_n, C_k) \cdot P(x_n \mid C_k) \cdot P(C_k)$$

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Here, we assume that x_1 doesn't depend on $x_2, x_3 \ldots x_n$.

$$P(C_k|x) = P(x_1|C_k) \cdot P(x_2|C_k) \ldots P(x_n|C_k) \cdot P(C_k)$$

$$P(C_k|x) = P(C_k) \cdot \prod_{i=1}^{n} P(x_i|C_k)$$

$\Sigma \rightarrow$ sum $\quad \Pi \rightarrow$ product.

Really, $P(C_k|x) \propto P(C_k) \cdot \prod_{i=1}^{n} P(x_i|C_k)$

coz we ignore denominator. $(P(x))$

Actually,

$$P(C_k|x) = \frac{1}{Z} P(C_k) \cdot \prod_{i=1}^{n} P(x_i|C_k)$$

$Z \rightarrow P(x)$

$$\hat{y} = \text{argmax} \; P(C_k) \cdot \prod_{i=1}^{n} P(x_i|C_k)$$

maximum posterior Rule.
calculate all and gives max value.
$k = \{1, 2, 3, \ldots, K\}$

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Training:

→ Make Lookup Table (Dictionary)
calculate all the probabilities and save in dictionary.

Numerical Data in Naive Bayes

Height	172	180	165	190	139	145	160	172
Weight	165	170	140	200	100	120	140	150
Gender	M	M	M	M	F	F	F	F

Now predict {H=185, W=170, G=?}.

- $P(M \mid H=185, W=170) = ?$

$P(H=185 \mid M) \cdot P(W=170 \mid M) \cdot P(M)$

In data there is no 185, hence $P(H=185 \mid M)=0$.
But, we have to assume 'Height is a Gaussian distributed random value' [normal].

1) First, calculate μ & σ

172 180 165 190

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Put in

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

this gives probability.

Do same for $P(W = 170 \mid F)$

Do same for ~~gen~~ Female.

- If distribution is Binomial → BinomialNB

  Multinomial NB
  Possion NB

* Use GaussianNB.

```
from sklearn.naive_bayes import GaussianNB
model = GaussianNB()
model.fit(X_train, y_train)
```